ETALE-LOCAL WELL-NIGH-AFFINE CHARTS FOR FINITE-INERTIA STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3797-3815 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3817-3851, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 31 of queue_stacks-stacks-more-morphisms.json (pb=1742, stmt 3797-3815, proof 3817-3851).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 03JE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3851–3851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-well-nigh-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3339–3346. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3851–3851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 050M: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3851–3851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CL0: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3851–3851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DU4: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3851–3851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DUA: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3851–3851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Assume $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is finite. Then there exist a set I and for $i \in I$ a morphism of algebraic stacks*

$$g_i : \mathcal{X}_i \longrightarrow \mathcal{X}$$

with the following properties

(1) $|\mathcal{X}| = \bigcup |g_i|(|\mathcal{X}_i|)$,
(2) *$\mathcal{X}_i$ is well-nigh affine,*
(3) *$\mathcal{I}_{\mathcal{X}_i} \to \mathcal{X}_i \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism, and*
(4) *$g_i : \mathcal{X}_i \to \mathcal{X}$ is representable by algebraic spaces, separated, and étale,*

Proof. For any $x \in |\mathcal{X}|$ we can choose $g : \mathcal{U} \to \mathcal{X}, \mathcal{U} = [U/R]$, and u as in Morphisms of Stacks, Lemma 0DU4. Then by Morphisms of Stacks, Lemma 0DUA we see that there exists an open substack $\mathcal{U}' \subset \mathcal{U}$ containing u such that $\mathcal{I}_{\mathcal{U}'} \to \mathcal{U}' \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism. Let $U' \subset U$ be the R-invariant open corresponding to the open substack $\mathcal{U}'$. Let $u' \in U'$ be a point of U' mapping to u. Observe that $t(s^{-1}(\{u'\}))$ is finite as $s : R \to U$ is finite. By Properties, Lemma 01ZY and Groupoids, Lemma 03JE we can find an R-invariant affine open $U'' \subset U'$ containing u'. Let R'' be the restriction of R to U''. Then $\mathcal{U}'' = [U''/R'']$ is an open substack of $\mathcal{U}'$ containing u, is well-nigh affine, $\mathcal{I}_{\mathcal{U}''} \to \mathcal{U}'' \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism, and $\mathcal{U}'' \to \mathcal{X}$ and is representable by algebraic spaces and étale. Finally, $\mathcal{U}'' \to \mathcal{X}$ is separated as $\mathcal{U}''$ is separated (Lemma lemma-well-nigh-affine) the diagonal of $\mathcal{X}$ is separated (Morphisms of Stacks, Lemma 0CL0) and separatedness follows from Morphisms of Stacks, Lemma 050M. Since the point $x \in |\mathcal{X}|$ is arbitrary the proof is complete. □